Total No. of Questions : 10]

SEAT No. :

P6527

[Total No. of Pages : 2

[5870]-2008

T.E. (Computer Engineering)
OPERATING SYSTEM DESIGN
(2012 Pattern) (Semester - I) (310242)

Time : 2½ Hours] **[Max. Marks : 70**

Instructions to the candidates:

1) ***Neat diagrams must be drawn wherever necessary.***
2) ***Figures to the right indicate full marks.***
3) ***Assume Suitable data, if necessary.***

Q1) a) What is Kernel? Explain Monolithic and Micro Kernel. **[5]**

b) Write a short note on OS booting process. **[5]**

OR

Q2) a) How hybrid system of Swapping and demand paging works? **[6]**

b) What is thread? How it is different than the process. **[4]**

Q3) a) A process references pages in the following order: 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7. Use LRU and FIFO page replacement algorithms to find out number of page faults for this reference string using 3 page frames. **[6]**

b) Explain Process State transitions with suitable diagram. **[4]**

OR

Q4) a) What is deadlock? What are the necessary conditions to occur deadlock? **[5]**

b) What is deadlock recovery? Explain *Resource Preemption* method to recover from deadlock. **[5]**

Q5) a) What is socket? What are different types of socket? Explain following system calls with parameters of socket mechanism. **[8]**

i) connect ii) listen iii) getsockname

b) Explain the problem of Multiprocessor Systems. How this problem is solved by using Master and Slave processors? **[8]**

OR

P.T.O.

Q6) a) What is Semaphore? How multiprocessor system is solved using Semaphore? **[8]**

b) Write a short note on Pipes. **[4]**

c) Write a short note on Tunis System. **[4]**

Q7) a) Write a short note on UEFI Boot and U-Boot. **[8]**

b) Explain Fedora-19 EFI files: *grubx64.efi, MokManager.efi*. **[8]**

OR

Q8) a) Write a note on Make Tools: make, nmake, cmake. **[8]**

b) Write a short note on AWK tool and Sorting tool. **[8]**

Q9) a) Write a short note on Google Android. **[6]**

b) What is Handheld operating system? How it is secured. **[6]**

c) Write a note on Linux Scheduling. **[6]**

OR

Q10) Write note on following: **[18]**

a) Palm OS.

b) Unix Free BSD Scheduling.

c) Microsoft Windows Mobile.

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